Gabriele Russo Russo

Curriculum Vitae

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Research Interests and Scientific Results

I am currently an Assistant Professor (*RTT*) at Tor Vergata University of Rome, where I received my PhD degree in 2021. My research interests are in the area of distributed computing systems, with particular emphasis on run-time performance management for distributed applications.

I co-authored 9 papers published in international journals, 26 papers published in international conference and workshop proceedings, and 2 peer-reviewed book chapters. As at August 2026, Google Scholar reports 858 citations of my papers, with a h-index equal to 16; Scopus reports 575 citations, with a h-index equal to 12.

Since November 2021, I am a member of the Review Board of *IEEE Transactions on Parallel and Distributed Systems*.

Education and Positions

Education

2017–21 **PhD student** in Computer Science, Tor Vergata University of Rome
Thesis: *Model-based Auto-Scaling of Distributed Data Stream Processing Applications.*
Advisors: Prof. Valeria Cardellini and Prof. Francesco Lo Presti.
Final examination grade: *Excellent.*

2019 **Visiting PhD student**, Imperial College London (UK)
Hosted by Dr. Giuliano Casale, August–September 2019.

2014–17 **Laurea Magistrale** (MSc equivalent) in Computer Engineering, Tor Vergata University of Rome, 110/110 *cum laude.*
Thesis: *Optimal Deployment and Run-Time Reconfiguration for Data Stream Processing.*
Advisors: Prof. Valeria Cardellini and Prof. Francesco Lo Presti.

2011–14 **Laurea** (BSc equivalent) in Computer Engineerinig, Tor Vergata University of Rome, 110/110 *cum laude.*
Thesis: *Analysis and Implementation of Energy-Aware Routing Algorithms for Ad-Hoc Wireless Networks.*
Advisor: Prof. Francesco Lo Presti.

Positions

2026– **Assistant Professor** (“RTT”) in Information processing systems (Italian SSD IINF- 05/A), Dept. of Civil Engineering and Computer Science Engineering, Tor Vergata University of Rome, since April 2026.

2023–26 **Research Fellow** (“RTDa”), Dept. of Civil Engineering and Computer Science Engineering, Tor Vergata University of Rome, April 2023–April 2026.

2023 **Research Assistant** (“Borsista”), Roma Tre University
Project: *Per una Giustizia Giusta: Innovazione ed Efficienza negli Uffici Giudiziari-Giustizia Agile.*
Coordinator: Stefano Iannucci.
Duration: 3 months (January–April).

2022–23 **Research Assistant** (“Assegnista di Ricerca”), Tor Vergata University of Rome
Project: *Earth In The Cloud, POR FESR Lazio 2014-2020.*
Coordinator: Francesco Quaglia.
Duration: 12 months.

2021–22 **Research Assistant** (“Assegnista di Ricerca”), Tor Vergata University of Rome
Project: *Self-adaptive Deployment of Data Stream Processing Applications in Edge and Serverless Environments.*
Coordinator: Valeria Cardellini.
Duration: 12 months.

Habilitations

2025 **Italian Scientific Habilitation (ASN)** for associate professorship, Italian SSD IINF-05/A

Honors, Awards and Grants

2024 *Google Cloud for Education*, Cloud credits (5,000 USD) for research on "Adaptive Resource Allocation for Serverless Functions".

2023 **Best Paper Award** in the Special Session on Compute Continuum at PDP ’23, with the paper [C15].

2019 7$^{\text{th}}$ Heidelberg Laureate Forum, selected for participation as a young researcher.

2011 **Alfiere del Lavoro**
Awarded by Italian President Giorgio Napolitano and *Federazione Nazionale Cavalieri del Lavoro*. I was included in the list of the 25 Italian students completing high school with the highest grades.

Research Projects

Responsibility

2025–27 *Development of communicating robotic IoT systems to be used in construction sites, roadworks, and quarries, for area delimitation and continuous collection of operational data*, **Work Package Leader**
Funded by INAIL within the BRIC program.
Duration: 24 months.
I am responsible for the WP4: *Design and Implementation of a Software Architecture for Data Processing.*

Participation

2023–25 Spoke 1 “Future HPC & Big Data” of *ICSC National Research Centre for High Performance Computing, Big Data and Quantum Computing*, **Research Scientist**
Funded by MUR Missione 4 Componente 2 Investimento 1.4: Potenziamento strutture di ricerca e creazione di “campioni nazionali” di R&S (M4C2-19) - Next Generation EU (NGEU).

2022 *Earth In The Cloud: Earth Observation in the Cloud*, **Research Scientist**
Funded by Regione Lazio – POR FESR Lazio 2014-2020 (Azione 1.2.1)
Local Project Leader: Francesco Quaglia.

Technology Transfer

2024–25 **Responsible** for the research activities on the *Design, implementation and evaluation of novel solutions for adaptive underwater systems and networks* funded by WSense s.r.l (`https://wsense.it/`)
Within a third-party outsourcing of research work agreement with the Dept. of Civil Engineering and Computer Science Engineering at Tor Vergata University of Rome.
Duration: 12 months .

Professional service

Editorial Roles

2026– Guest Editor for a special issue of *Future Generation Computer Systems* (Elsevier) on "Lightweight and Sustainable Computing across the Cloud-Edge Continuum", along with V. Cardellini, A. Ding, L. Bittencourt, M. Nardelli.

2024–25 Guest Editor for a special issue of *ACM Transactions on Autonomous and Adaptive Systems* on "Artificial Intelligence for Adaptive and Autonomous Cloud/Edge Computing Systems", along with V. Cardellini, I. Dusparic, S. Iannucci: `https://dl.acm.org/doi/10.1145/3796726`

Conference and Workshop Organization

Workshop Co-Chair

- *4th International Workshop on Scalable Compute Continuum (WSCC 2026)*, co-located with Euro-Par 2026, Pisa (Italy), August, 2026.
- *4th International Workshop on Artificial Intelligence for Autonomous computing Systems (AI4AS 2025)*, co-located with IEEE ACSOS 2026, Cesena (Italy), September, 2026.
- *International Workshop on Agentic AI and LLMs in the Computing Continuum (AI-CC 2025)*, co-located with IEEE/ACM UCC 2025, Nantes (France), December, 2025.
- *International Workshop on Intelligent and Scalable Systems across the Computing Continuum (ScaleSys 2025)*, co-located with IoT 2025, Wien (Austria), November 2025.
- *3rd International Workshop on Artificial Intelligence for Autonomous computing Systems (AI4AS 2025)*, co-located with IEEE ACSOS 2025, Tokyo (Japan), September, 2025.
- *2nd International Workshop on Artificial Intelligence for Autonomous computing Systems (AI4AS 2024)*, co-located with IEEE ACSOS 2024, Aarhus (Denmark), September 20, 2024.
- *International Workshop on Quality of Service-aware Serverless Computing (QServ 2023)*, co-located with IEEE/ACM UCC 2023, Taormina (Italy), December 4, 2023.
- *First International Workshop on Artificial Intelligence for Autonomous computing Systems (AI4AS 2023)*, co-located with IEEE ACSOS 2023, Toronto (Canada), September 28, 2023.

TPC Member

- *IEEE International Conference on Distributed Computing Systems* (ICDCS 2025, ICDCS 2026)
- *IEEE International Parallel & Distributed Processing Symposium* (IPDPS 2026)
- *International European Conference on Parallel and Distributed Computing* (EuroPar 2026)
- *IEEE International Symposium on Cluster, Cloud, and Internet Computing* (CCGRID 2025, CCGRID 2026)
- *International Conference on Cloud Computing and Services Science* (CLOSER 2025, CLOSER 2026)

- *IEEE Real-Time and Embedded Technology and Applications Symposium* (RTAS 2024, RTAS 2026)
- *ACM/SIGAPP Symposium On Applied Computing* (SAC 2025, SAC 2026)
- *ACM International Conference on Distributed and Event-based Systems* (DEBS 2022, DEBS 2025, DEBS 2026)
- *International Conference on Computing, Networking and Communications* (ICNC 2025)
- *Workshop on Artificial Intelligence for Performance Modeling, Prediction, and Control* (AIPerf 2024, AIPerf 2025)
- *International Workshop on Scalable Compute Continuum* (WSCC 2023, WSCC 2024, WSCC 2025)
- *International Workshop on the Orchestration of the Serverless Edge-Cloud Continuum* (COHERENT 2024)
- *International Workshop on Sustainable Service-Oriented Computing: Addressing Environmental, Social, and Economic Dimensions* (SSCOPE 2023)
- TPC Chair, *4th International Workshop on Self-Protecting Systems* (SPS 2022)
- Shadow TPC Member, *16th ACM EuroSys* (EuroSys 2021)

Tutorials co-Chair

- *7th IEEE International Conference on Autonomic Computing and Self-Organizing Systems (ACSOS 2026)*

Publicity & Social Media Chair

- *30th IEEE Real-Time and Embedded Technology and Applications Symposium (RTAS 2024)*
- *14th ACM/SPEC International Conference on Performance Engineering (ICPE 2023)*

Reviewer for Journals and Conferences

I reviewed manuscripts submitted to the following international journals:

- Pervasive and Mobile Computing, Elsevier (2026)
- Performance Evaluation, Elsevier (2026)
- IEEE Transactions on Parallel and Distributed Systems (2021–2026)
- ACM Transactions on Autonomous and Adaptive Systems (2024–2026)
- IEEE Transactions on Network and Service Management (2022, 2026)
- Cluster Computing, Springer (2019–2023, 2026)
- IEEE Transactions on Cloud Computing (2022, 2025)
- Journal of Grid Computing, Springer (2020–2022, 2025)
- IEEE Transactions on Computers (2024)
- Future Generation Computer Systems, Elsevier (2019, 2023–2024)
- Journal of Systems and Software, Elsevier (2023)
- Parallel Computing, Elsevier (2022–2023)
- ACM Computing Surveys (2022)
- IEEE Transactions on Mobile Computing (2022)
- Journal of Parallel and Distributed Computing, Elsevier (2022)
- IEEE Internet of Things Journal (2021,2022)
- Information Systems, Elsevier (2021)
- Science of Computer Programming, Elsevier (2020)

- Expert Systems with Applications, Elsevier (2018)

Since November 2021, I am a member of the ***Review Board*** **of IEEE Transactions on Parallel and Distributed Systems**.

I also served as an external reviewer for the following international conferences:
- IEEE International Conference on Cloud Engineering (IC2E) (2025)
- ACM Distributed and Event-based Systems (2019,2022)
- ACM/SPEC International Conference on Performance Engineering (2021)
- IEEE/ACM International Conference on Utility and Cloud Computing (2020,2021)
- IEEE Vehicular Technology Conference (2018)

Invited Talks and Tutorials

Invited Talks

- "Serverless Functions across the Cloud- Edge Continuum: Challenges and Perspectives", invited talk at the *2nd Workshop on Accelerated HPC in the Cloud-Edge Continuum*, co-located with IEEE IC2E '25, Rennes, France, September 23, 2025.
- "The Impact of AI on Data Centers" (in Italian), invited talk at *RomeCup 2024*, Rome, Italy, March 21, 2024.
- "Using Reinforcement Learning to Control Auto-Scaling of Distributed Applications", invited talk at the *1st Workshop on Artificial Intelligence for Performance Modeling, Prediction, and Control*, co-located with ACM/SPEC ICPE '23, Coimbra, Portugal, April 15, 2023.
- "Meet-the-author: Serverledge: Decentralized Function-as-a-Service for the Edge-Cloud Continuum", invited talk at the *2nd Workshop on Serverless computing for pervasive cloud-edge-device systems and services*, co-located with IEEE Percom '23, Atlanta, USA, March 13, 2023.

Conference Tutorials

- F. Lo Presti, G. Russo Russo, V. Cardellini, "Reinforcement learning for run-time performance management in the Cloud/Edge", *39th International Symposium on Computer Per- formance, Modeling, Measurements and Evaluation 2021 (Performance 2021)*, Milan, Italy, 8-12 November, 2021.

Teaching

I have been involved in the following teaching activities (at Tor Vergata University of Rome, if not explicitly stated):

Doctoral Courses

- *Reinforcement Learning for Run-Time Performance Management* (10 hours), Tor Vergata University of Rome, March–April 2026.
- *Reinforcement Learning for Run-Time Performance Management* (10 hours), Tor Vergata University of Rome, March–April 2024.
- *Model-free and Model-based Auto-Scaling Techniques for Distributed Applications* (10 hours), Roma Tre University, June 2023.

Graduate and Master Degree Courses

- *Machine Learning* (9 ECTS), for the MSc in *Computer Engineering* (2025/26).
- *Machine Learning* (9 ECTS; 6 ECTS taught by me), for the MSc in *Computer Engineering* (2023/24, 2024/25).
- *Distributed Systems and Cloud Computing* (9 ECTS; 1 ECTS taught by me), for the MSc in *Computer Engineering* (2023/24, 2024/25).
- *Hands-on Cloud Computing Services*, supplementary course (10 hours) for the MSc in *Computer Engineering* (from 2019/20 to 2022/23).
- *Hands-on Big Data & Hadoop* (12 hours), 2nd Level Master degree in *Customer Experience & Social Media Analytics (CESMA) (from 2018/19 to 2025/26).*
- *Hands-on Big Data* (16 hours), 1st Level Master degree in *Data Science (2021/22,2022/23).*

Undergraduate Courses

- *Fundamentals of Programming and Data Analytics* (3 ECTS), BSc course at the Department of Engineering, Roma Tre University. (2022/23).
- *MIPS Assembler Programming*, supplementary course (10 hours) for the BSc in *Computer Engineering* (2021/22).
- TA for the *Computer Architecture* course, BSc in *Computer Engineering* (from 2016/17 to 2020/21).

PhD Student Supervision

2023– Cecilia Calavaro, expected 2027, co-supervised with V. Cardellini.
2025– Simone Nicosanti, expected 2029, co-supervised with V. Cardellini.
2025– Emanuele Valzano, expected 2029, co-supervised with V. Cardellini.

Thesis Supervision

Since 2023, I have supervised 22 *Laurea Magistrale* (MSc) theses and, since 2018, I have been assistant supervisor of 18 theses at Tor Vergata University of Rome.

Publications

International Journals

J1 C. Calavaro, V. Cardellini, F. Lo Presti, and G. Russo Russo. Carbon-aware offloading with function variants for serverless computing in the cloud-to-edge continuum. *Future Gener. Comput. Syst.*, 185:108670, 2026. URL: `https://doi.org/10.1016/j.future.2026.108670`, `doi:10.1016/J.FUTURE.2026.108670`

J2 C. Calavaro, V. Cardellini, F. Lo Presti, and G. Russo Russo. Beyond cloud: Serverless functions in the compute continuum. *SN Computer Science*, 6(194):1–19, 2025. `doi:10.1007/s42979-025-03699-7`

J3 G. Russo Russo, D. Ferrarelli, D. Pasquali, V. Cardellini, and F. Lo Presti. Qos-aware offloading policies for serverless functions in the cloud-to-edge continuum. *Future Generation Computer Systems*, 156:1–15, 2024. `doi:10.1016/j.future.2024.02.019`

J4 G. Russo Russo, V. Cardellini, and F. Lo Presti. A framework for offloading and migration of serverless functions in the edge–cloud continuum. *Pervasive and Mobile Computing*, 100(101915):1–15, 2024. `doi:10.1016/j.pmcj.2024.101915`

J5 G. Russo Russo, V. Cardellini, and F. Lo Presti. Hierarchical auto-scaling policies for data stream processing on heterogeneous resources. *ACM Trans. Auton. Adapt. Syst.*, 18(4):14:1–14:44, 2023. `doi:10.1145/3597435`

J6 V. Cardellini, F. Lo Presti, M. Nardelli, and G. Russo Russo. Run-time adaptation of data stream processing systems: The state of the art. *ACM Computing Surveys*, 54(11s):237:1–237:36 (plus 20–page suppl. material), 2022. `doi:10.1145/3514496`

J7 G. Russo Russo, M. Nardelli, V. Cardellini, and F. Lo Presti. Multi-level elasticity for wide-area data streaming systems: A reinforcement learning approach. *Algorithms*, 11(9):134:1–134:27, 2018. `doi:10.3390/a11090134`

J8 V. Cardellini, F. Lo Presti, M. Nardelli, and G. Russo Russo. Decentralized self-adaptation for elastic data stream processing. *Future Generation Computer Systems*, 87:171–185, 2018. `doi:10.1016/j.future.2018.05.025`

J9 V. Cardellini, F. Lo Presti, M. Nardelli, and G. Russo Russo. Optimal operator deployment and replication for elastic distributed data stream processing. *Concurrency and Computation: Practice & Experience*, 30(9):e4334:1–e4334:20, 2018. `doi:10.1002/cpe.4334`

International Conferences and Workshops

C1 A. Finocchi, G. Marseglia, S. Corsi, G. Russo Russo, and G. Bianchi. Toward a combined view of intrusion and evasion: an adversary-aware IDS. In *Proceedings of the IEEE International Black Seas Conference on Communications and Networking (BlackSeaCom '26)*, pages 1–4, 2026. URL: `https://doi.org/10.1109/BlackSeaCom69760.2026.11598180`, `doi:10.1109/BLACKSEACOM69760.2026.11598180`

C2 M. Lupini, G. Russo Russo, and V. Cardellini. Spec2faas: using ai agents for autonomous serverless function development and deployment. In *Proceedings of the 20th ACM International Conference on Distributed and Event-based Systems, DEBS 2026, Lisbon, Portugal, June 23-26, 2026*, 2026. `doi:10.1145/3809481.3812613`

C3 Simone Nicosanti, Gabriele Russo Russo, and Valeria Cardellini. Energy- and quantization-aware DNN partitioning in the edge-cloud continuum (work in progress paper). In *Proceedings of the 17th ACM/SPEC International Conference on Performance Engineering, ICPE Companion 2026*, pages 47–54. ACM, 2026. `doi:10.1145/3777911.3801106`

C4 Romolo Marotta, Gabriele Russo Russo, Francesco Quaglia, and Pierangelo Di Sanzo. A bootstrapping technique for reducing the costs of machine learning models for predicting execution times in iaas clouds. In *Proceedings of the 2025 ACM Symposium on Cloud Computing (SoCC '25)*, SoCC '25, page 790–802, New York, NY, USA, 2026. ACM. `doi:10.1145/3772052.3772251`

C5 E. Incerto, R. Pizziol, G. Russo Russo, and M. Tribastone. Wasteless: An optimal provisioner for self-adaptive second-generation serverless applications. In *Proceedings of the 20th International Conference on Software Engineering for Adaptive and Self-Managing Systems (SEAMS), Ottawa, ON, Canada, April 28-29, 2025*, pages 61–72, Los Alamitos, CA, USA, 2025. IEEE Computer Society. `doi:10.1109/SEAMS66627.2025.00015`

C6 F. Righetti, B. Cornacchia, G. Russo Russo, N. Tonellotto, V. Cardellini, and C. Vallati. Energy-efficient function invocation scheduling for edge faas platforms. In *Proceedings of the 2025 IEEE International Conference on Smart Computing (SMARTCOMP)*, pages 18–25, 2025. `doi:10.1109/SMARTCOMP65954.2025.00057`

C7 G. Russo Russo, P. Spaziani, and V. Cardellini. Towards QoS-aware serverless function offloading in the edge-cloud continuum through reinforcement learning. In *Proceedings of 2025 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*, pages 1073–1080, New York, NY, USA, 2025. IEEE. `doi:10.1109/IPDPSW66978.2025.00168`

C8 G. Russo Russo, E. D'Alessandro, V. Cardellini, and F. Lo Presti. Towards a multi-armed bandit approach for adaptive load balancing in function-as-a-service systems. In *IEEE International Conference on Autonomic Computing and Self-Organizing Systems, ACSOS 2024 - Companion*, pages 103–108, 2024. `doi:10.1109/ACSOS-C63493.2024.00039`

C9 C. Calavaro, G. Russo Russo, M. Salvati, V. Cardellini, and F. Lo Presti. Towards energy-aware execution and offloading of serverless functions. In *Proceedings of the 4th Workshop on Flexible Resource and Application Management on the Edge, FRAME 2024, Pisa, Italy, June 3-7, 2024*, pages 23–30. ACM, 2024. `doi:10.1145/3659994.3660313`

C10 M. Nardelli and G. Russo Russo. Function offloading and data migration for stateful serverless edge computing. In *Proceedings of the 15th ACM/SPEC International Conference on Performance Engineering, ICPE 2024, London, United Kingdom, May 7-11, 2024*, pages 247–257, New York, NY, USA, 2024. ACM. `doi:10.1145/3629526.3649293`

C11 A. Semjonov, G. Russo Russo, H. Bornholdt, and J. Edinger. Wasimoff: Distributed computation offloading using webassembly in the browser. In *Proceedings of 2024 IEEE International Conference on Pervasive Computing and Communications Workshops and other Affiliated Events, PerCom 2024 Workshops, Biarritz, France, March 11-15, 2024*, pages 2023–208. IEEE, 2024

C12 M. Nardelli, G. Russo Russo, and V. Cardellini. Compute continuum: What lies ahead? In *Euro-Par 2023: Parallel Processing Workshops - Euro-Par 2023 International Workshops, Limassol, Cyprus, August 28 - September 1, 2023, Revised Selected Papers, Part I*, volume 14351 of *Lecture Notes in Computer Science*, pages 5–17. Springer, 2023. `doi:10.1007/978-3-031-50684-0\_1`

C13 F. Filippini, R. Cavadini, D. Ardagna, R. Lancellotti, G. Russo Russo, V. Cardellini, and F. Lo Presti. Figaro: reinforcement learning management across the computing continuum. In *3rd International Workshop on Distributed Machine Learning for the Intelligent Computing Continuum (DML-ICC 2023), Published in: Proceedings of the IEEE/ACM 16th International Conference on Utility and Cloud Computing, UCC 2023, Taormina (Messina), Italy, December 4-7, 2023*, number 47, pages 1–8. ACM, 2023

C14 G. Russo Russo, T. Mannucci, V. Cardellini, and F. Lo Presti. Artifact: Serverledge: Decentralized function-as-a-service for the edge-cloud continuum. In *Proceedings of IEEE International Conference on Pervasive Computing and Communications Workshops and other Affiliated Events, PerCom Workshops 2023, Atlanta, GA, USA, March 13-17, 2023*, pages 1–2. IEEE, 2023. `doi:10.1109/PerComWorkshops56833.2023.10150345`

C15 G. Russo Russo, V. Cardellini, and F. Lo Presti. Serverless functions in the cloud-edge continuum: Challenges and opportunities. In *Proceedings of the 31st Euromicro International Conference on Parallel, Distributed and Network-based Processing, PDP 2023, Naples, Italy, March 1-3, 2023*, pages 321–328. IEEE, 2023. `doi:10.1109/PDP59025.2023.00056`

C16 C. Calavaro, G. Russo Russo, and V. Cardellini. Real-time analysis of market data leveraging apache flink. In *Proceedings of the 16th ACM International Conference on Distributed and Event-based Systems, DEBS 2022, Copenhagen, Denmark, June 27 - 30, 2022*, pages 162–165, 2022. `doi:10.1145/3524860.3539650`

C17 G. Russo Russo, A. Milani, S. Iannucci, and V. Cardellini. Towards QoS-aware function composition scheduling in apache openwhisk. In *Proceedings of 2022 IEEE International Conference on Pervasive Computing and Communications Workshops and other Affiliated Events, PerCom 2022 Workshops, Pisa, Italy, March 21-25, 2022*, pages 693–698. IEEE, 2022. `doi:10.1109/PerComWorkshops53856.2022.9767299`

C18 G. Russo Russo, V. Cardellini, G. Casale, and F. Lo Presti. Mead: Model-based vertical auto-scaling for data stream processing. In *Proceedings of 21th IEEE/ACM International Symposium on Cluster, Cloud and Internet Computing, CCGRID '21, Virtual Event, May 10-13, 2021*, pages 314–323, Los Alamitos, CA, USA, 2021. IEEE Computer Society. `doi:10.1109/CCGrid51090.2021.00041`

C19 G. Russo Russo, A. Schiazza, and V. Cardellini. Elastic pulsar functions for distributed stream processing. In *ICPE '21: ACM/SPEC International Conference on Performance Engineering, Virtual Event, France, April 19-21, 2021*, pages 9–16. ACM, 2021. `doi:10.1145/3447545.3451901`

C20 G. Russo Russo. Model-based auto-scaling of distributed data stream processing applications. In *Proceedings of the 21st International Middleware Conference Doctoral Symposium, Middleware 2020, Virtual Event / Delft, The Netherlands, December 07-11, 2020*, pages 5–8. ACM, 2020. `doi:10.1145/3429351.3431741`

C21 G. Russo Russo. Self-adaptive data stream processing in geo-distributed computing environments. In *Proceedings of the 13th ACM International Conference on Distributed and Event-based Systems, DEBS 2019, Darmstadt, Germany, June 24-28, 2019*, pages 276–279. ACM, 2019. `doi:10.1145/3328905.3332304`

C22 G. Russo Russo, V. Cardellini, and F. Lo Presti. Reinforcement learning based policies for elastic stream processing on heterogeneous resources. In *Proceedings of the 13th ACM International Conference on Distributed and Event-based Systems, DEBS 2019, Darmstadt, Germany, June 24-28, 2019*, pages 31–42, 2019. `doi:10.1145/3328905.3329506`

C23 M. Nardelli, G. Russo Russo, V. Cardellini, and F. Lo Presti. A multi-level elasticity framework for distributed data stream processing. In *Euro-Par 2018: Parallel Processing Workshops - Euro-Par 2018 International Workshops, Turin, Italy, August 27-28, 2018, Revised Selected Papers*, volume 11339 of *Lecture Notes in Computer Science*, pages 53–64. Springer, 2018. `doi:10.1007/978-3-030-10549-5\_5`

C24 G. Russo Russo. Towards decentralized auto-scaling policies for data stream processing applications. In *Proceedings of the 10th Central European Workshop on Services and their Composition, Dresden, Germany, February 8-9, 2018*, volume 2072 of *CEUR Workshop Proceedings*, pages 47–54. CEUR-WS.org, 2018. URL: `http://ceur-ws.org/Vol-2072/paper8.pdf`

C25 V. Cardellini, F. Lo Presti, M. Nardelli, and G. Russo Russo. Towards hierarchical autonomous control for elastic data stream processing in the fog. In *Euro-Par 2017: Parallel Processing Workshops - Euro-Par 2017 International Workshops, Santiago de Compostela, Spain, August 28-29, 2017, Revised Selected Papers*, volume 10659 of *Lecture Notes in Computer Science*, pages 106–117. Springer, 2017. `doi:10.1007/978-3-319-75178-8\_9`

C26 V. Cardellini, F. Lo Presti, M. Nardelli, and G. Russo Russo. Auto-scaling in data stream processing applications: A model-based reinforcement learning approach. In *New Frontiers in Quantitative Methods in Informatics - 7th Workshop, InfQ 2017, Venice, Italy, December 4, 2017, Revised Selected Papers*, volume 825 of *Communications in Computer and Information Science*, pages 97–110. Springer, 2017. `doi:10.1007/978-3-319-91632-3\_8`

Book Chapters

B1 A. Alnafessah, G. Russo Russo, V. Cardellini, G. Casale, and F. Lo Presti. Ai-driven performance management in data-intensive applications. In N. Zincir-Heywood, Y. Diao., and M. Mellia, editors, *Communications Network and Service Management in the Era of Artificial Intelligence and Machine Learning*. Wiley, 2021. `doi:10.1002/9781119675525.ch9`

B2 G. Russo Russo, V. Cardellini, F. Lo Presti, and M. Nardelli. Towards a security-aware deployment of data streaming applications in fog computing. In Wei Chang and Jie Wu, editors, *Fog/Edge Computing For Security, Privacy, and Applications*, pages 355–385. Springer International Publishing, Cham, 2021. `doi:10.1007/978-3-030-57328-7_14`

PhD Thesis

G. Russo Russo. *Model-based Auto-Scaling of Distributed Data Stream Processing Applications*. PhD thesis, University of Rome Tor Vergata, 2021

Software

- **Serverledge**, `https://github.com/serverledge-faas/serverledge`
  I am the original author and main contributor of Serverledge, an open-source Function-as-a-Service framework designed for Edge-Cloud environments. Serverledge relies on a decentralized architecture, with no centralized gateways for function invocations, and supports horizontal and vertical function execution offloading.

Rome, August 21, 2026